

Apply Proportional Reasoning in Graphs, Finance, and Modelling

Step 11

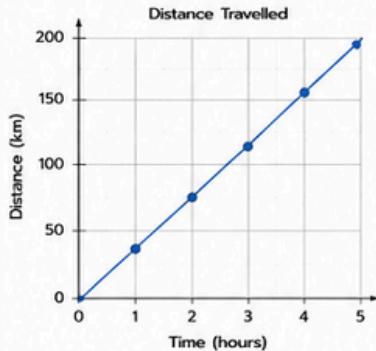
Elementary
(Grades 2–4)



Today I am learning to use proportional reasoning to solve problems using graphs, finance, and models.

1. Read the Graph

The graph shows the distance travelled by a car at a constant speed. Use the graph to answer the questions.



- How far does the car travel in 1 hour? _____ km
- How far in 3 hours? _____ km
- How long does it take to travel 160 km? _____ h
- How far will the car travel in 2.5 hours? _____ km
- What is the speed of the car in km per hour? _____

2. Finance – Simple Interest

Simple interest is paid on the original amount of money. Use the table to answer the questions.

	Amount Invested (\$)	Interest Rate per Year	Time (years)	Interest Earned (\$)	Total Amount (\$)
a)	1,000	5%	1	_____	_____
b)	1,000	5%	3	_____	_____
c)	2,500	4%	2	_____	_____
d)	800	7.5%	4	_____	_____



Remember!

Interest = Principal × Rate × Time
Total Amount = Principal + Interest

3. Finance – Best Value

Three phone plans are shown below. Use proportional reasoning to decide which plan is the best value. Show your working.

Plan A

\$18 for 1 GB of data

Plan B

\$36 for 3 GB of data

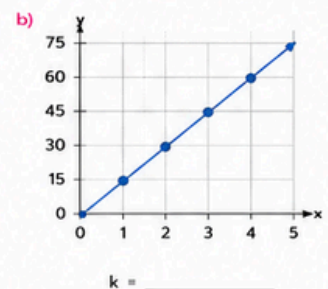
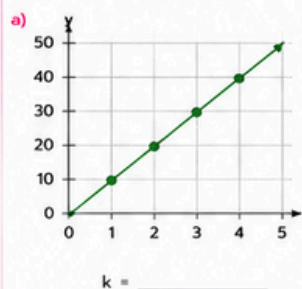
Plan C

\$45 for 5 GB of data

- Find the cost per GB for each plan.
Plan A: \$_____ per GB Plan B: \$_____ per GB Plan C: \$_____ per GB
- Which plan is the best value? _____
Why? _____

4. Graphs – Proportional Relationships

Each graph shows a proportional relationship. Write the constant of proportionality (k) for each.



5. Modelling – Real-Life Situation

A recipe says 2 cups of flour are needed for every 3 cups of sugar. Use proportional reasoning to complete the table.

Flour (cups)	2	4	6	8	10
Sugar (cups)	3				



- How much sugar is needed for 8 cups of flour? _____ cups
- How much flour is needed for 12 cups of sugar? _____ cups
- Write the ratio of flour to sugar in simplest form. _____

7. Modelling – Scale Drawing

A park is drawn on a scale of 1 cm = 10 m. Use the scale to find the real length.



- If a path on the plan is 6.4 cm long, what is the real length? _____ m
- If the real length of another path is 85 m, what is the length on the plan? _____ cm

6. Finance – Discounts

An item has a price of \$120. Calculate the sale price after each discount. Show your working.

a) 10% off	b) 25% off	c) 30% off
Sale price: \$ _____	Sale price: \$ _____	Sale price: \$ _____

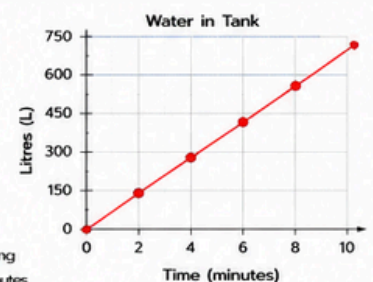


Tip!
To find a percentage of an amount:
Amount × Percentage (as a decimal)
For example: 25% of \$120 = \$120 × 0.25 = \$30

8. Challenge Problem

A water tank fills at a constant rate. The graph shows the amount of water in the tank over time.

- How many litres of water are in the tank after 6 minutes? _____ L
- How long will it take to fill 600 L? _____ minutes
- If the tank already has 150 L, how long will it take to be full (600 L)? _____ minutes



I can...

- ☐ read and interpret graphs.
- ☐ solve problems using simple interest and discounts.
- ☐ compare unit rates and best value.
- ☐ use proportional reasoning in real-life models.
- ☐ explain my thinking.

Key Vocabulary

- proportional** – having a constant ratio between quantities
- constant of proportionality** – the multiplier (k) in $y = kx$
- unit rate** – a rate for one unit
- interest** – extra money earned on an investment
- discount** – a reduction from the original price
- scale** – a ratio that shows how a drawing compares to real life



Great job!

You are using proportional reasoning to make smart decisions in graphs, finance, and modelling!